JIAQI SHI

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Education

Columbia University — New York, NY
Ph.D. in Operations Research — Aug 2022 - Present
M.S. in Operations Research — Aug 2020 - May 2022
Advisor: Adam Elmachtoub

Nanjing University — Nanjing, CN
B.S. in Mathematics — Sep 2016 - Jun 2020

University of California, Berkeley — Berkeley, CA
Exchange Student — Jan 2019 - May 2019

Research Interests

Revenue Management and Pricing, Data-Driven Decision Making, Market Design

Publications and Preprints

1. Adam Elmachtoub, Jiaqi Shi. Tight Guarantees for Static Policies in Vehicle Sharing Networks.
   - Vehicle-sharing networks are prevalent in bike, scooter, car sharing, and ride-sharing services. Establish tight universal guarantees for static policies and efficiently compute the optimal static policy.

2. Omar El Housni, Adam Elmachtoub, Harsh Sheth, Jiaqi Shi. Price and Assortment Optimization under the Multinomial Logit Model with Opaque Products. Major revision at Management Science.
   The 21st Conference on Web and Internet Economics (WINE), 2025.
   - An opaque product is one for which only partial information is disclosed to the buyer at the time of purchase. Integrate opaque products into the MNL model and solve the associated price and assortment optimization problems.

3. Adam Elmachtoub, Jiaqi Shi. The Power of Static Pricing for Reusable Resources. Major revision at Management Science.
   The 26th ACM Conference on Economics and Computation (EC), 2025.
   - Finalist, INFORMS Service Science Best Student Paper Award, 2025
   - Reusable resources are used in many real-world applications such as cloud computing, car rental, and sharing economy. Provide universal high-performance guarantees for pricing reusable resources using a simple static policy under various model settings.

4. Jiaqi Shi, Ginger Y. Ke, Zizhuo Wang, Lianmin Zhang. Price Optimization for a Multistage Choice Model.
   Production and Operations Management, 33(7), 1535-1551, 2024.
   - Propose a multi-stage choice model where a customer's past purchase choices could influence future prices that she can receive. Show that a unique optimal pricing policy exists under mild conditions.

Industry Experience

Amazon — Bellevue, WA
Applied Scientist Intern — Jun 2025 - Aug 2025
Project: Assortment Optimization under Tight Fulfillment Center Capacity Constraints.

Talks

Price and Assortment Optimization under the Multinomial Logit Model with Opaque Products

- The 21st Conference on Web and Internet Economics, New Brunswick, NJ — December 2025

- INFORMS Annual Meeting, Atlanta, GA — October 2025
- INFORMS RMP Conference, New York, NY — July 2025

The Power of Static Pricing for Reusable Resources

- INFORMS Annual Meeting, San Francisco, CA — November 2026
- INFORMS RMP Conference, Ann Arbor, MI — July 2026
- INFORMS MSOM Conference, Boston, MA — July 2026
- The 26th ACM Conference on Economics and Computation (EC'25), Stanford, CA — July 2025
- INFORMS RMP Conference, Los Angeles, CA — July 2024

Honors and Awards

- Finalist, INFORMS 2025 Service Science Best Student Paper Award — 2025
- Cheung-Kong Innovation Doctoral Fellow — 2024-2026
- Finalist, INFORMS 2023 BSS Data Challenge Competition — 2023
  - Report: Decoupling Price and Demand Estimation for Profit Optimization
- Columbia IEOR Department Graduate Fellowship — 2022

Teaching Experience

Teaching Assistant

- IEOR 4650 Business Analytics — Spring 2024
- IEOR 4004 Optimization Models and Methods — Fall 2023
- IEOR 4601 Dynamic Pricing and Revenue Management — Spring 2023

Course Assistant

- IEOR 4418 Transportation Analytics and Logistics — Spring 2022
- IEOR 4106 Stochastic Models — Fall 2021

Professional Service

Reviewer

- Journal referee for *Operations Research*, *Operations Research Letters*, *Transportation Research Part E*

Department Service

- Inaugural Member of IEOR Ph.D. Council — Oct 2022 - May 2025

Skills

- Technical: Python | MATLAB | Gurobi | LaTeX
- Languages: English (Fluent), Mandarin (Native)

References

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